

TRACKING SPIES IN THE SKIES



ABOUT THE TALK

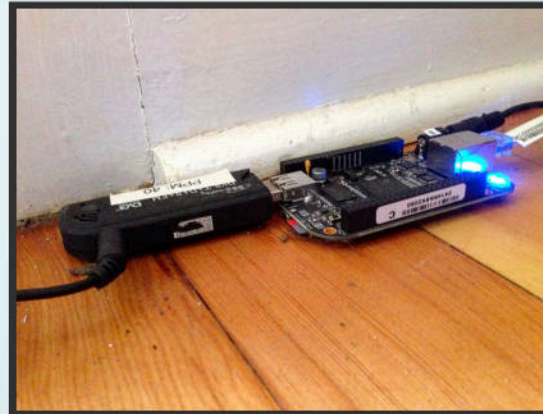
LAW ENFORCEMENT AND AERIAL SURVEILLANCE

- History of aerial surveillance (Sam Richards)
- Technology on spy planes (Jerod MacDonald-Evoy)
- Detecting surveillance aircraft (Jason Hernandez)



HISTORY OF THE SKY SPIES

- Odd plane patterns noticed, WSJ, Baltimore
- r/conspiracy (John Wiesman - ADSB Detection)
- Citizen journalists (Sam Richards) #FBISkySpies and 100 Tail-numbers, links to FlightRadar24 tracks



SKY SPIES 101

- Sam's story goes viral, a week later AP breaks it into the mainstream
- Sen. Franken calls for investigation (nothing happens)
- FBI Planes hidden behind front companies (FVX Research, et. al)



WHAT WE KNOW



TYPES OF AIRCRAFT

- Small fixed wing (Cessnas)
- Large dual engine (Beechcraft)
- Military style (Pilatus)
- Helicopters
- Drones (Small and Large)



EQUIPMENT

- Wescam by L3 Communications
- FLIR SAFIRE
- IMSI Catchers
- LETC Devices

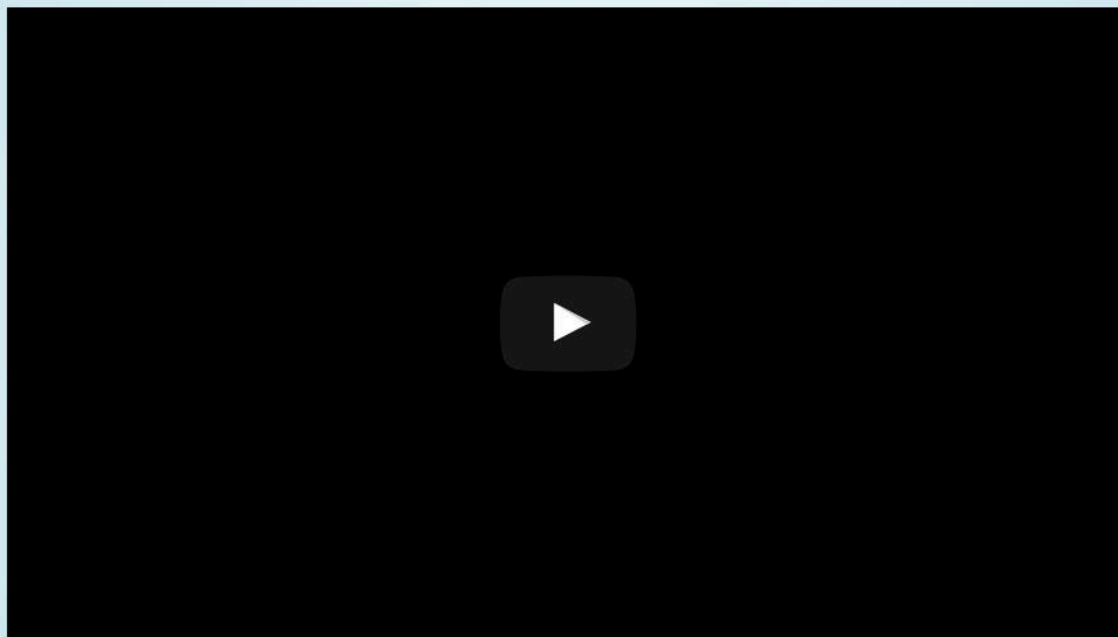


EXAMPLES OF USE

- FBI Aerial Surveillance of Freddie Grey protests
- FBI Aerial Surveillance of Arizona I-10 shooter suspect's apartment
- Phoenix PD used Pilatus to follow U-haul thief
- 'Persistent Surveillance Solutions'



PHOENIX PD PLANE



FOOTAGE OBTAINED VIA PUBLIC RECORDS REQUEST

TRACKING THE SKY SPIES

- How do we more generally detect surveillance aircraft and activity?
- Registrations can be changed and obscured
- Many surveillance technologies are commercially available
- How much surveillance is happening in other parts of the world?
- Technical and operational requirements dictate flight patterns
- Surveillance flights look very different from most other traffic

HOW DO WE TRACK AIRCRAFT?

- Radar is not practical
- ADS-B messages are the way to go
- Active community of radio / aviation / hacking enthusiasts collect ADS-B data
- Requires a Raspberry Pi 1B+, an RTL-SDR radio, antenna, and internet connection
- Multiple aggregators collect data
 - FlightRadar24.com, FlightAware.com, **adsbexchange.com**
- FAA regulations require an increasing number of aircraft to transmit ADS-B
 - Part of the "NextGen" program
- Similar regulations in .EU, .IN, .AU, elsewhere

ADS-B DATA

- Aircraft transmit a beacon signal with a unique ICAO number
- Positions can be calculated with multilateration
 - Compare time difference of messages arriving at multiple receivers
- Requires 4+ receivers for accurate calculation
- Aggregator networks collect feeds from ADS-B receivers and calculate aircraft positions
- Some aircraft also transmit additional information: (latitude / longitude), call sign, etc.
 - Currently not required, and location may not be accurate

LIMITATIONS TO DATA

- Major commercial flight tracking sites augment their data with FAA radar data
- This data comes with restrictions that tracking sites do not publish positions of aircraft on the FAA's ASDI block list
- Bulk access to data is limited or expensive
- ADS-B Exchange is an exception
- Does not use FAA data, does not censor flights
- Provides free access to live & historical data
 - Data challenges
 - Donation info on their site

PICKING SURVEILLANCE FLIGHTS FROM A FIREHOSE OF DATA

- There are over 80,000 flights a day
- At any given time 8,000~13,000 aircraft are in the air
- Most of these are not surveillance flights
- How do we pick out the surveillance flights?

SURVEILLANCE FLIGHTS VS. OTHERS

- Most non-surveillance traffic goes from point A to B as quickly and directly as possible
- Minimizes flying over populated areas and crossing in to airports' controlled airspace
- Exceptions - holding patterns, flight schools, aerial surveys

TECHNICAL CONSTRAINTS OF SURVEILLANCE FLIGHTS

- Technical and economic constraints result in relatively unique flight patterns for surveillance
- Cell site simulators - range of ~2 miles
- FLIR (infrared) cameras
- Surveillance flights often take off and land at the same airport
- Cover densely populated metro areas
- Visual surveillance - needs daylight
- Electronic surveillance - cover of night preferred
- Altitude "sweet spot"

PATTERN BASED DETECTION

- Surveillance flights make a large number of turns
- Most flights with 30+ turns "look" like surveillance flights
- Limitations & future improvement



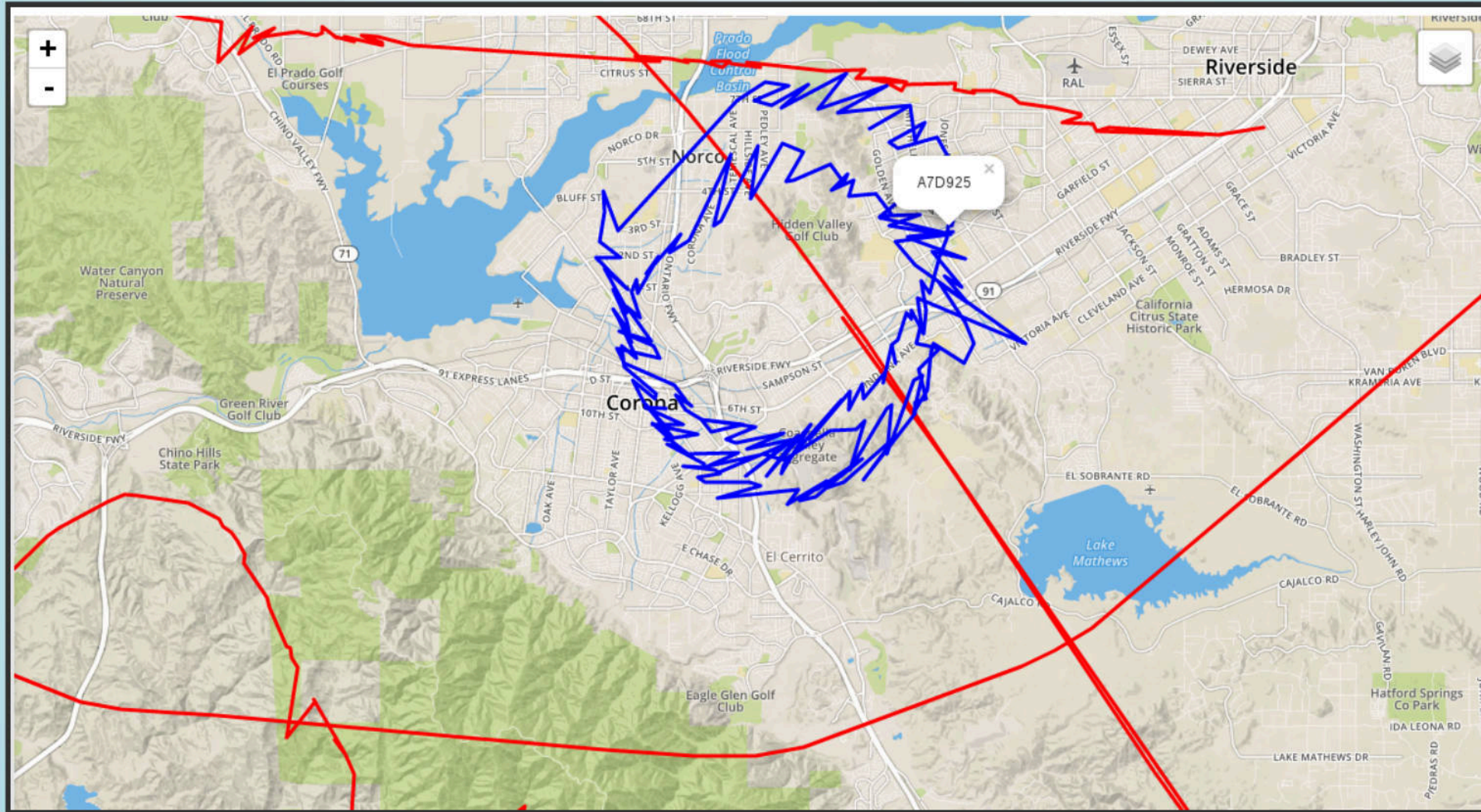
SURVEILLANCE SCORE METHODOLOGY

- Calculate headings of each aircraft and increase the score each time it changes > 90 degrees
- Conditional based on altitude
 - Sweet spot is appx. 6,000 - 12,000 ft
- Future refinements:
 - Consider proximity to airports and controlled airspace (needs good airspace data, may be compute intensive)
 - Score based on aircraft model
 - Additional geometric calculations to filter out survey activity
 - Compare flights to interesting geography -- borders, events, etc.

IMPLEMENTATION

- Virtual Radar Server (<http://www.virtualradarserver.co.uk>) with connection to adsbexchange.com 's live data feed
- Analysis / tracking code to be released today pulls flight trails from local Virtual Radar Server JSON endpoint
- Flight data queued in RabbitMQ and composed in Redis
- Uses multiple cores and flight analysis can be distributed to multiple machines
 - Completed flights stored for retrospective analysis
- Flight paths for each suspicious aircraft exported as JSON files
- Upload to object storage (AWS S3)
 - Viewable in a basic leaflet.js web map

EXAMPLE



CONCLUSION

- Many surveillance technologies improve with Moore's Law
- Policies and oversight have not moved as quickly
- You can work on tracking spy planes
- Use, fork, and improve our application
- Set up your own receiver and feed to adsbexchange.com and any future open ADS-B aggregators

MORE INFO:

- <https://www.nstarpost.com>
- github.com/nstarpost
- twitter.com/nstarpost

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